N83 guarded factor-base continuation

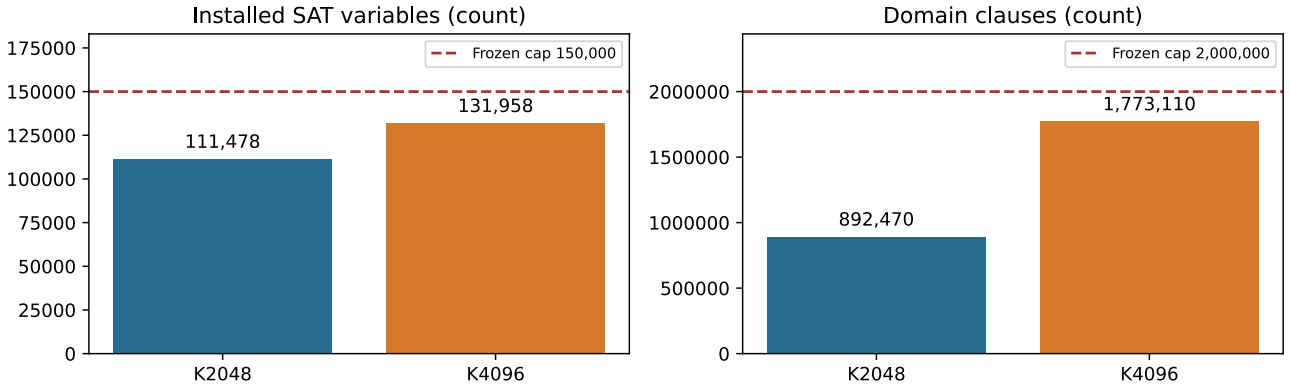
18 October 2026 · cloud-matched-k2048-k4096-20261018

Both models constructed within frozen variable/domain-clause limits. No complete cold solve time, factor-base winner or speed improvement is established.

Frozen source: main protocol f08f9f887ba9f87d0b8976ec1c7a6cd65eb63127; short control 19c2f73268be9c4fa16b18f55f234e077ffb5e53. Native Rust worker, historical Python supervisor and auditor. Updated native-only policy requires native replacements before further comparisons.

Method: public synthetic N83 Koblitz fixtures; m=5 orbit-phase chained S3; K2048 then K4096; one trial each, 32-model maximum. Native x86_64 cloud VM; Docker 1 CPU, 4 GiB, zero swap, network disabled, read-only inputs. Quotas verified; timing isolation and A/A controls not established. Archives fetched and verified from setec-astronomy-artifacts.

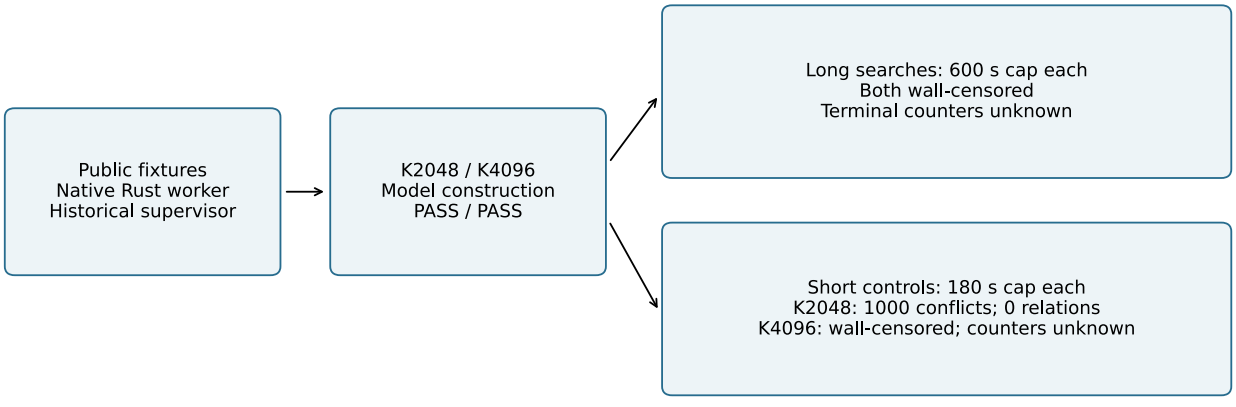
N83 guarded model construction | n=1 per arm; exact recorded counts



cloud-matched-k2048-k4096-20261018 | no uncertainty interval recorded; not a speed comparison

Counts measured once per arm, no interval recorded. The 2,000,000 limit is on domain clauses, not total clauses. Capacity outer wall: K2048 23314.246807 ms; K4096 12539.253286 ms. These are diagnostic process measurements, not throughput comparisons.

Observed execution and terminal evidence | one trial per arm



Identical fixture files; paired terminal query identity unverified for K4096.

Outcomes, custody and limits

Long searches (100,000-conflict / 600 s limits) ended at outer wall 600612.228016 ms and 601214.695177 ms. Final solver counters are unknown in both.

The separate 1,000-conflict / 180 s controls ended at 162630.491881 ms (K2048) and 181095.333452 ms (K4096). K2048 emitted exactly 1000 conflicts, 1 SAT call, 0 models, 0 relations and rank 0; independent original-group replay passed. K4096 was wall-censored, with final counters unknown. Coarse start events do not establish SAT entry. Identical fixture files do not establish paired terminal query equality for K4096.

All six stage audits passed within their complete/censored scopes. Six forged-result controls were rejected. Frozen-source tests and post-merge integration tests apply to different source revisions; see the integration addendum. No further scientific comparisons were run after the native-only policy arrived.

Evidence: repository-relative links below resolve from the Markdown report beside this PDF. The source of truth is:

`research/koblitz_n83_factor_base_sweep_20261008/verification/cloud-matched-k2048-k4096-20261018/`

Read PROTOCOL.md, SHORT_CONTROL_PROTOCOL.json, summary.json, query-identity.json, accounting.json, VALIDATION.md and raw-artifact-manifest.json there. Figure data is copied directly from recorded summary counts; no new research computation. REPORT.md carries clickable evidence links and editable rendering sources.

Custody: both factor bases are stored and GET/hash-verified in setec-astronomy-artifacts. Historical old-bucket sidecars are preserved unchanged. The broader 5,771,044-byte worker/source/log ZIP remains local because automatic approval review blocked upload; no ZIP upload is claimed.

Remaining scope: the full finite-grid sweep and historical total-cost ledger remain partial. Next discriminating work requires a native supervisor/auditor, native K4096 phase markers and a public positive control. The independent emulated K4096 result is a separate host/binary and cannot supply missing native counters.

Canonical graphs checked and unchanged: docs/ic-leaderboard.html, docs/index-calculus-scoreboard.html and docs/ic/progress-timeline.json. This bounded diagnostic establishes no qualified gain or full cold winner. Local capacity figures show the newly recorded facts; benchmark copies are diagnostic only.